

Pocket Labs Science Kit

WHAT'S IN THE KIT

- 3 PocketLab Voyagers (motion, magnetic, temp, pressure sensors)
- 1 scale
- 1 100-ft measuring tape
- 1 compass
- 2 analog clocks



QUICK-START GUIDE

1. Connect a Voyager to a device via Bluetooth using the PocketLab app.
2. Pick a sensor stream (acceleration, altitude, magnetic field...).
3. Attach it to whatever's moving — cart, ball, shoe, sled.
4. Record, graph, and export the data live.

FULL LESSONS IN THIS GUIDE

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PRO TIP: The PocketLab app graphs in real time — project it so the whole class sees data appear as it happens.

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LESSON · GRADES 3-5

Playground Physics Hunt

LEARNING OUTCOMES

- Students will collect acceleration data using a wireless sensor
- Students will make and test predictions about motion
- Students will read a live graph and connect spikes to physical events

MATERIALS

- PocketLab Voyagers (3 rotation stations)
- Tablets/Chromebooks with the PocketLab app
- Playground access (slide, swing, ball)
- Prediction sheet

INSTRUCTIONS

1. Indoors demo: shake the sensor and watch the live graph spike — connect motion to line.
2. Students predict: which playground spot makes the biggest spike?
3. Stations rotate: attach the Voyager to slide, swing, and a rolled ball.
4. At each station, one student moves, others watch the graph and mark the peak.
5. Back inside, compare peaks against predictions.
6. Circle: why did the winner win? What does 'acceleration' seem to mean?

ACTIVITY

Playground Physics Hunt — real data from their own bodies in motion. Watching the graph spike at the exact moment of the swing's turnaround is the intuition textbooks can't deliver.

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LESSON · GRADES 6-8

Crash Test Engineering

LEARNING OUTCOMES

- Students will design a crash-protection device to reduce measured impact
- Students will use sensor data to evaluate design performance
- Students will apply an evidence-based iteration cycle

MATERIALS

- PocketLab Voyagers mounted on toy carts or rolling platforms
- Cardboard, tape, rubber bands, foam scraps for bumpers
- Ramp and wall crash site
- Impact data table

INSTRUCTIONS

1. Baseline: crash the bare cart three times; record peak acceleration each run.
2. Teams design and build a bumper within a size limit.
3. Crash trials: three runs per design; record and average the peaks.
4. Iterate once with a documented design change.
5. Leaderboard by percent reduction from baseline, not raw numbers.
6. Debrief: which physical strategies (crumple, deflect, cushion) showed up in the data?

ACTIVITY

Crash Test Engineering — build bumpers, crash carts, and compete on measured impact reduction. Averaging multiple runs sneaks in measurement reliability alongside the crunching.

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LESSON · GRADES 9-12

Motion Graph Matching

LEARNING OUTCOMES

- Students will reproduce a target motion graph through physical movement
- Students will interpret velocity and acceleration graphs precisely
- Students will design graph puzzles that isolate specific kinematic concepts

MATERIALS

- PocketLab Voyagers (1 per group)
- Devices with the PocketLab app
- Printed target graphs (constant velocity, acceleration, direction change)
- Clear hallway or lab space

INSTRUCTIONS

1. Groups get target graph #1 and must move the sensor to recreate its shape.
2. Compare live capture to target; diagnose mismatches in kinematic language.
3. Progress through harder targets: steady speed, speeding up, stop-and-reverse.
4. Flip it: each group authors one target graph puzzle for another group.
5. Author groups judge whether solvers truly matched, defending their ruling.
6. Exit: sketch the velocity graph for a described motion, unaided.

ACTIVITY

Motion Graph Matching — become the graph. Physically recreating a velocity curve, then authoring puzzles for rivals, builds kinematics intuition that transfers straight to exam problems.